

Ashwin Sateesh Kumar

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EXPERIENCE

AI Software Engineer

Dec. 2024 – Present

Citibank

Dallas, TX

- Engineered a ReAct-based multi-agent vulnerability detection system with dependency-aware code analysis using LangGraph and AST parsing, improving detection accuracy by 25%
- Automated vulnerability analysis, exploit simulation, and remediation workflows by developing a report-generation agent into the ReAct pipeline, reducing manual penetration testing effort by 50%+
- Architected multi-agent infrastructure on OpenShift — FastAPI and LiteLLM gateway with circuit breakers and fallback chain, PostgreSQL + Redis state management, and HPA auto-scaling, serving 8,000+ analysts
- Designed LangGraph two-loop orchestration — supervisor LLM routing between parallel sub-agents with scoped MCP tools, mid-request checkpointing, and Tekton CI/Harness CD canary deployment
- Optimized a RAG pipeline with BM25 + cross-encoder reranking and built OTel/Prometheus observability with RAGAS evaluation on production traces, improving retrieval precision by 25% and faithfulness above 0.95

Machine Learning Engineer

Jan. 2024 – Dec. 2024

Abecedarian

Boston, MA

- Designed a multimodal AI assistant using LangChain with queue-based continual learning and a web data ingestion pipeline for fine-tuning GPT-3.5, enabling real-time model updates validated via perplexity score
- Reduced image inference latency by 2× by distilling Stable Diffusion into a lightweight latent diffusion model using U-Net pruning while maintaining CLIP alignment scores

Research Assistant

July 2023 – Nov. 2023

Northeastern University

Boston, MA

- Uncovered demographic bias (visual and linguistic) in 400k social media posts by training a disentangled multimodal β -VAE with a Mixture of Experts (MoE) fusion layer over U-Net and BERT encoders
- Reduced inference latency by 30% on GCP via bfloat16 quantization, enabling counterfactual fairness auditing by manipulating disentangled latent factors across demographic groups

Machine Learning Engineer Intern, R&D

June 2022 – Dec. 2022

Signify (Philips Lighting)

Boston, MA

- Engineered a real-time user re-identification system using omni-scale feature learning, achieving 0.95 mAP in identity matching across multi-camera views, deployed for enterprise smart lighting optimization

Software Engineer

July 2019 – Nov. 2020

KPIT Technologies Ltd.

Bengaluru, India

- Developed an autonomous driving perception system using U-Net scene segmentation and vision-radar sensor fusion, achieving 0.89 IoU on real-time traffic object detection

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, R

Frameworks: PyTorch, TensorFlow, Scikit-Learn, LangChain, LangGraph, FastAPI, LiteLLM, React

Infrastructure: Git, Bash, Docker, OpenShift, Tekton, Harness, Redis, PostgreSQL, Prometheus, Grafana

Cloud: GCP (Vertex AI, Cloud Storage), AWS (S3, SageMaker)

Libraries: NumPy, pandas, SciPy, OpenCV, NLTK, XGBoost, FAISS, HF Transformers, RAGAS, OpenTelemetry

PROJECTS

HealthBot | *Python, PyTorch, Flask, HTML, Docker*

Sep. 2023 – Dec. 2023

- Developed a medical chatbot by fine-tuning FLAN-T5 with LoRA and RLHF (PPO), achieving 0.75 semantic similarity on clinical responses and 0.96 F1 on disease classification via BERT-based named entity recognition

EDUCATION

Northeastern University

Boston, MA

Master of Science in Data Science (GPA: 3.8/4.0)

Sep. 2021 – Dec. 2023

PES University

Bengaluru, India

Bachelor of Engineering in Electronics and Communications

Aug. 2015 – May 2019